Secure Network Design for Matadors University

A Comprehensive Enterprise Network Architecture Proposal

IS457 – Advanced Telecommunications and Networking

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Executive Summary

Matadors University, an institution comprising eight academic colleges with a combined faculty of 642 members, requires a secure, segmented, and scalable enterprise network capable of supporting approximately 2,568 connected devices within a strict capital budget of $200,000. This report presents a complete network design that addresses these requirements through a three-tier hierarchical architecture combining VLAN-based segmentation, dynamic redundancy mechanisms, layered security controls, and integrated wireless coverage.

The proposed design assigns each of the eight colleges its own VLAN and IP subnet drawn from the private 10.0.0.0/16 address space. Seven colleges receive a /23 subnet capable of accommodating up to 510 host addresses, while the smallest college, Health and Human Sciences, uses a /24 subnet for greater address efficiency. Inter-VLAN routing is performed by a Cisco ISR 4331 router configured with eight Dot1Q sub-interfaces, an arrangement known as router-on-a-stick. Two Cisco Catalyst 2960-48TT-L switches serve as distribution-layer aggregators, interconnected by a static EtherChannel link and configured as the primary and backup spanning tree root bridges using Rapid Per-VLAN Spanning Tree Protocol (RSTP). Sixty-three additional Catalyst 2960 switches form the access layer, each dual-homed to both distribution switches for failover resilience.

Security is implemented as a defense-in-depth strategy across four layers: a Cisco ASA 5506-X firewall at the perimeter handling network address translation and stateful inspection; extended access control lists on the router enforcing inter-VLAN traffic policies; switch-level port security limiting MAC addresses per access port and triggering automatic shutdown on violations; and device hardening on every Cisco device including SSH-only access, encrypted password storage, login banners, and disabled unused ports. Wireless coverage is provided by twenty-five Cisco Aironet 3702i access points connected to college-specific VLANs, ensuring that wireless traffic inherits the same segmentation and security policies as wired traffic. Mobile devices receive addressing automatically via DHCP pools hosted on the central router, while fixed devices retain static IP assignments as specified by the project brief.

The total budget for the proposed design is $142,616, leaving a substantial contingency margin of $57,384 that may be reserved for unforeseen costs or used for discretionary enhancements during deployment. The design has been validated through a Cisco Packet Tracer simulation covering sixteen test scenarios including intra-VLAN connectivity, inter-VLAN routing, ACL enforcement, spanning tree convergence, EtherChannel link aggregation, port security, DHCP attribution, and wireless connectivity. All sixteen tests produced expected results, confirming that the design satisfies the segmentation, redundancy, security, and operational requirements of the project brief.

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1. Introduction

The challenge presented by Matadors University is representative of those faced by mid-sized academic institutions transitioning from legacy flat networks to modern segmented enterprise architectures. With eight colleges of varying sizes, ranging from the College of Humanities at 119 faculty members to the College of Health and Human Sciences at 45, the institution requires a network capable of providing isolated environments for each academic unit while still allowing controlled communication between them. Each faculty member is assumed to operate four primary devices: a desktop personal computer, a laptop, an Internet Protocol telephone, and a shared network printer. This assumption yields a total of approximately 2,568 active network devices that the design must accommodate, plus future growth.

The project brief imposes several constraints that have shaped the design decisions throughout this proposal. The total budget is capped at $200,000, encompassing all hardware, cabling, professional services, and first-year maintenance. The brief mandates static Internet Protocol (IP) addressing, which simplifies certain operational tasks but limits flexibility for mobile devices; this report addresses that limitation through a hybrid scheme that retains static addressing for fixed equipment while introducing Dynamic Host Configuration Protocol (DHCP) services for wireless clients. The design must be implementable in Cisco Packet Tracer, which restricts the device models available and influences how some configurations are demonstrated. Finally, all costs must be traceable to verifiable vendor sources, which has guided the selection of equipment toward Cisco-certified models with stable list pricing.

This report is organized as follows. Section 2 presents the key assumptions underlying the design. Section 3 describes the network topology, including the three-tier hierarchical architecture and the rationale for each layer. Section 4 details the IP addressing scheme and VLAN structure. Section 5 covers routing, spanning tree, EtherChannel, and security configurations. Section 6 documents the wireless infrastructure and DHCP services. Section 7 reports on the testing and troubleshooting performed during simulation. Section 8 presents the itemized budget, and Section 9 concludes with reflections on the design and recommendations for future enhancements.

2. Key Assumptions

The following assumptions form the foundation of the design and have been documented explicitly to clarify the scope of the proposal and the boundaries within which the recommendations apply. First, each faculty member is assumed to operate four network-connected devices: a desktop computer, a laptop, an IP telephone, and shared access to a networked printer. This assumption derives directly from the project brief and yields the total device count of 2,568. Second, the campus is assumed to occupy a single physical site, eliminating the need for wide area network (WAN) links between distant locations. Third, the access layer is assumed to be deployed using stacked Cisco Catalyst 2960 switches, with each college's switch stack sized in proportion to its faculty count. The College of Humanities, with 476 devices, requires eleven 48-port switches; the College of Health and Human Sciences, with 180 devices, requires five.

Fourth, the network design uses a hybrid IP addressing strategy. Static IP addresses are assigned to fixed devices, namely desktop computers, IP telephones, network printers, and access points, while a dynamic DHCP pool provides addresses to mobile wireless clients including laptops, smartphones, and tablets. This approach satisfies the brief's static addressing requirement for fixed equipment while acknowledging the operational impracticality of manually configuring IP addresses on every mobile device that enters the campus. Fifth, end-user computing equipment is procured under separate IT budgets controlled by each individual college and is therefore excluded from the network infrastructure budget presented in this report. Including 642 desktop computers and 642 laptops would have required approximately $1.2 million in additional spending, far exceeding the $200,000 allocation. Sixth, the design assumes a five-to-seven-year refresh cycle for network infrastructure, which is consistent with industry practice and aligns with the depreciation schedules typically applied to enterprise networking equipment.

3. Network Topology and Architecture

The network design follows a three-tier hierarchical model widely adopted in enterprise campus deployments. This model separates network functions into clearly distinguished layers, each with a specific role and a corresponding set of devices, simplifying operations, troubleshooting, and future scaling.

3.1. Three-Tier Hierarchy

At the topmost layer, an Internet uplink terminates at a Cisco ASA 5506-X next-generation firewall. The firewall provides network address translation (NAT) for outbound traffic, stateful inspection of inbound and outbound flows, and a controlled boundary between the campus network and external networks. Behind the firewall, a Cisco ISR 4331 Integrated Services Router serves as the routing core. This router performs inter-VLAN routing through eight logical sub-interfaces created on top of a single physical Gigabit Ethernet port. Each sub-interface is associated with one VLAN through the IEEE 802.1Q encapsulation protocol and acts as the default gateway for that VLAN's subnet. This arrangement is referred to as router-on-a-stick and is the standard approach when a Layer 3 switch is not used or available.

The distribution layer comprises two Cisco Catalyst 2960-48TT-L switches, designated Dist-SW-1 and Dist-SW-2. These switches aggregate traffic from the access layer and provide the redundancy backbone of the design. They are interconnected by a static EtherChannel link consisting of two Gigabit Ethernet cables bundled into a single logical link, providing both increased aggregate bandwidth and tolerance to single-link failures. RSTP is configured to elect Dist-SW-1 as the root bridge, with priority 4096, while Dist-SW-2 is configured with priority 8192 to serve as the backup root.

The access layer comprises 63 Cisco Catalyst 2960-48TT-L switches distributed across the eight colleges. Each switch provides 48 FastEthernet ports for end-device connectivity plus two Gigabit uplinks. These uplinks are configured as trunk ports and connected to both Dist-SW-1 and Dist-SW-2, an arrangement known as dual-homing. RSTP elects one of the two uplinks as the active forwarding path while placing the other in a blocking state, preventing forwarding loops. In the event that the active uplink or its corresponding distribution switch fails, RSTP reconverges in approximately one second and activates the backup uplink, restoring connectivity without manual intervention.

3.2. Topology Diagram Description

The complete topology can be visualized as follows. At the top of the diagram, a cloud icon represents the Internet, connected by a single line to the ASA 5506-X firewall. The firewall connects to the ISR 4331 router via a dedicated Gigabit Ethernet link addressed in the 192.168.100.0/30 subnet. The router's primary interface, GigabitEthernet 0/0/1, connects to Dist-SW-1 through a trunked link carrying all eight VLANs. Dist-SW-1 and Dist-SW-2 are interconnected by the EtherChannel bundle. Below the distribution layer, eight access switch groups are shown, one per college, each labeled with its VLAN identifier (VLAN 10 for Humanities, VLAN 20 for Business, and so forth through VLAN 80 for Health and Human Sciences). Each access switch group connects upward to both Dist-SW-1 and Dist-SW-2 through dual trunk links. At the lowest level of the diagram, end devices including PCs, IP phones, printers, and access points are shown attached to the access switches.

4. IP Addressing Scheme and VLAN Configuration

4.1. Address Block Selection

The design uses the private address block 10.0.0.0/16 as defined in RFC 1918. This block provides 65,534 usable host addresses, which is more than sufficient for the projected 2,568 devices and leaves significant headroom for institutional growth, future VLAN additions, and unanticipated infrastructure such as a guest wireless network or an isolated server VLAN. The use of a private block also requires only a single public IP address at the firewall for outbound translation, which is both cost-effective and operationally simple.

4.2. Subnet Sizing

Within the parent block, each college receives a contiguous subnet sized to accommodate its device count plus reasonable growth headroom. Seven of the eight colleges are assigned a /23 subnet, which yields 510 usable host addresses. This is sufficient for all colleges except for very large ones, and even the College of Humanities with 476 devices fits within a /23 with approximately 7% growth margin. The College of Health and Human Sciences, with only 180 devices, is assigned a /24 subnet of 254 usable hosts, providing 41% headroom in a more compact address footprint.

The selection of /23 over /24 for most colleges deserves explanation. A /24 subnet of 254 hosts would not accommodate the 476 devices of the College of Humanities, the 408 of Business and Economics, or the 368 of Social Sciences. While it would have been possible to assign two /24 subnets to each large college, this would have required additional VLANs, additional inter-VLAN routing entries, and increased configuration complexity for marginal benefit. The /23 approach simplifies operations and provides a single subnet per college, aligning with the brief's requirement of one subnet per college.

4.3. VLAN-to-Subnet Mapping

Table 1 presents the complete VLAN-to-subnet mapping. VLAN identifiers are spaced in increments of ten to leave room for additional VLANs in the future. Each subnet is aligned to the appropriate boundary; specifically, /23 subnets begin on even-numbered third octets to ensure correct subnet calculation. The default gateway for each subnet is set to the last usable IP address, a convention adopted from the IS457 lecture material and consistent with enterprise practice.

Table 1: VLAN-to-subnet mapping for the eight colleges.

VLAN	College	Subnet	Gateway	Devices
10	Humanities	10.0.0.0/23	10.0.1.254	476
20	Business & Economics	10.0.2.0/23	10.0.3.254	408
30	Social Sciences	10.0.4.0/23	10.0.5.254	368
40	Education	10.0.6.0/23	10.0.7.254	320
50	Science & Mathematics	10.0.8.0/23	10.0.9.254	280
60	Engineering	10.0.10.0/23	10.0.11.254	276
70	Art & Media	10.0.12.0/23	10.0.13.254	260
80	Health & Human Sci.	10.0.14.0/24	10.0.14.254	180

4.4. Static and Dynamic Address Ranges

Within each subnet, the address space is partitioned into a static range and a DHCP range. The static range, comprising IP addresses ending in .1 through .99, is reserved for fixed devices: PCs, IP phones, network printers, and access points. These devices benefit from predictable addressing for asset tracking, network management, and troubleshooting. The DHCP range, comprising IP addresses from .100 to the broadcast minus two, is allocated dynamically to mobile wireless clients including laptops, smartphones, and tablets. This partitioning preserves the static-addressing requirement of the brief for fixed equipment while introducing operational flexibility for mobile devices.

4.5. VLAN Configuration on Access Switches

Each access switch is configured with all eight VLANs in its VLAN database, even though only one of these VLANs is actively assigned to its access ports. This deliberate choice ensures that the trunk database remains consistent across the campus, simplifies configuration management through a uniform template, and allows for easy reassignment of access ports if a faculty member temporarily relocates between colleges. This approach mirrors the propagation behavior of VLAN Trunking Protocol (VTP) in production enterprise networks, where VLAN information is maintained consistently across all participating switches.

5. Routing, Spanning Tree, EtherChannel, and Security

5.1. Inter-VLAN Routing

Inter-VLAN routing is performed by the ISR 4331 router using a router-on-a-stick configuration. The router's GigabitEthernet 0/0/1 interface is configured as a trunk on the switch side, and on the router side, eight sub-interfaces are created, one per VLAN. Each sub-interface is configured with the IEEE 802.1Q encapsulation directive specifying its VLAN identifier, and is assigned the gateway IP address for that VLAN's subnet. When a host in VLAN 10 sends a packet destined for a host in VLAN 60, for example, the packet is forwarded by the access switch up the trunk to the router, where it arrives on sub-interface 0/0/1.10. The router consults its routing table, identifies that the destination network 10.0.10.0/23 is reachable via sub-interface 0/0/1.60, applies any relevant access control lists, and forwards the packet down the trunk toward its destination.

5.2. Rapid Spanning Tree Protocol

To prevent broadcast loops in the redundant topology while maintaining sub-second convergence times, the design uses Rapid Per-VLAN Spanning Tree Protocol (RSTP) with the Cisco extension Per-VLAN Spanning Tree (PVST). RSTP elects one switch as the root bridge for each VLAN. In this design, Dist-SW-1 is configured with priority 4096 to ensure it is elected as the root bridge for all VLANs. Dist-SW-2 is configured with priority 8192, making it the second-best candidate and ensuring it will be elected as the root in the event that Dist-SW-1 fails.

For each access switch, RSTP calculates the lowest-cost path to the root bridge. With a Gigabit Ethernet uplink to Dist-SW-1 and a parallel Gigabit Ethernet uplink to Dist-SW-2, the direct path to Dist-SW-1 has a cost of 4 while the indirect path via Dist-SW-2 and the EtherChannel has a cost of 8. Consequently, RSTP designates the uplink to Dist-SW-1 as the active forwarding port and places the uplink to Dist-SW-2 in a blocking state. PortFast is enabled on all access ports to skip the listening and learning phases of STP, allowing end devices to obtain network connectivity immediately upon connection.

5.3. EtherChannel

The two Gigabit Ethernet links between Dist-SW-1 and Dist-SW-2 are bundled into a single logical link using EtherChannel, configured in static mode with the channel-group 1 mode on directive on both switches. This produces a 2 Gbps logical link that is presented to RSTP as a single port, eliminating the loop that would otherwise exist with two parallel links and avoiding the unnecessary blocking of one of them. The EtherChannel also provides resilience against single-

cable failures: if one of the two physical cables fails, traffic continues to flow over the remaining cable without any reconvergence event at the spanning tree layer.

5.4. Security Architecture

Security is implemented through four distinct layers, each addressing a different category of threats. At the perimeter, the ASA 5506-X firewall provides network address translation, stateful packet inspection, and policy-based filtering of traffic between the campus and the Internet. At the network layer, extended access control lists (ACLs) on the router enforce inter-VLAN traffic policies. As a representative example, an ACL named BLOCK_HUM_ENG denies traffic between the College of Humanities (10.0.0.0/23) and the College of Engineering (10.0.10.0/23) while permitting all other inter-VLAN flows. This type of policy could be applied to enforce specific institutional rules, such as preventing certain colleges from accessing each other's research servers without explicit authorization.

At the switch layer, port security is configured on all access ports. The configuration limits each port to a maximum of two MAC addresses (accommodating, for example, a PC and an IP phone connected through a daisy-chain configuration), with sticky MAC learning enabled for automatic provisioning, and a violation policy that automatically shuts down the port if an unauthorized device attempts to connect. This protects against rogue device insertion, MAC flooding attacks, and unauthorized network access through unattended jacks.

At the device layer, all Cisco devices are hardened. Telnet is disabled in favor of SSH for encrypted remote access. The service password-encryption directive applies a basic cipher to all stored passwords. Strong privileged-mode passwords are configured using the enable secret command, which uses MD5 hashing rather than the reversible encoding used by enable password. A login banner is displayed before authentication, providing a legal warning against unauthorized access. Unused physical ports are administratively shut down to prevent rogue device insertion.

6. Wireless Infrastructure and DHCP Services

6.1. Wireless Coverage

To support mobile faculty devices, the design incorporates 25 Cisco Aironet 3702i lightweight access points distributed across the eight colleges. The number of access points was sized using a target ratio of approximately one AP per 25 to 30 simultaneous wireless users, a guideline consistent with enterprise Wi-Fi deployment best practices in academic environments. Each access point is connected to its college's access switch on a dedicated FastEthernet port (specifically Fa0/4), which is configured as a standard access port assigned to the college's VLAN.

This integration allows wireless devices to inherit the same VLAN segmentation as wired devices: a laptop connected to the wireless network at the College of Humanities is placed in VLAN 10 by the access switch, just as a wired desktop in the same college would be. Consequently, all the security policies enforced through inter-VLAN ACLs apply identically to wireless and wired traffic, simplifying policy management and ensuring consistent behavior across connection types.

Each access point broadcasts a unique service set identifier (SSID) corresponding to its college, such as Matadors-Humanities, Matadors-Business, and so forth. All SSIDs use WPA2-PSK authentication with AES encryption, providing strong cryptographic protection of wireless traffic. A future enhancement, discussed in Section 9, would migrate to WPA2-Enterprise with 802.1X authentication, replacing the shared pre-shared key with per-user credentials authenticated against a central directory service.

6.2. DHCP Services

The router hosts eight DHCP pools, one per VLAN, each configured to allocate addresses from the dynamic range of its corresponding subnet. The static range from .1 to .99 is excluded from each pool through the ip dhcp excluded-address directive, ensuring that statically assigned addresses are never inadvertently distributed by DHCP. The DHCP pools provide each client with an IP address, a subnet mask, the appropriate default gateway, and a public DNS server address. Lease durations are set to seven days, balancing operational stability against efficient address reclamation.

The placement of DHCP services on the router rather than on dedicated DHCP servers simplifies the design by consolidating routing and addressing functions on a single device. Because each sub-interface acts simultaneously as a default gateway and as a DHCP relay endpoint for its VLAN, no explicit DHCP relay configuration (ip helper-address) is required. This is advantageous in a deployment of this scale, where introducing additional servers would add complexity and procurement overhead without commensurate benefit.

7. Testing and Troubleshooting

The design has been validated through a comprehensive Cisco Packet Tracer simulation comprising sixteen distinct test scenarios. Each test was designed to verify a specific aspect of the design and to confirm that the corresponding configuration produces the expected behavior.

The first set of tests verifies basic connectivity. Test 1 confirms that two devices in the same VLAN can ping each other directly through their access switch. Test 2 verifies that inter-VLAN routing functions correctly by sending an ICMP echo request from the College of Humanities to the College of Education and receiving a successful reply via the router. Test 3 verifies that the BLOCK_HUM_ENG access control list correctly blocks traffic between the College of Humanities and the College of Engineering, with the ping request producing the expected "Request timed out" response.

The second set of tests verifies the operational state of switching protocols. Test 4 confirms that all eight VLANs are correctly defined and assigned to ports through the show vlan brief command. Test 5 verifies that the trunk links carry all eight VLANs through the show interfaces trunk command. Test 6 verifies that Dist-SW-1 is correctly elected as the spanning tree root bridge by examining the output of show spanning-tree vlan 10, where the bridge identifier matches the root identifier. Test 7 verifies the EtherChannel state through show etherchannel summary, which reports Po1 in the SU state, indicating a Layer 2 channel that is up and using its underlying interfaces. Test 8 confirms that port security is active and configured correctly through show port-security.

The third set of tests verifies routing functionality. Test 9 confirms through show ip interface brief that all eight router sub-interfaces are in the up/up state, and through show ip route that the routing table contains eight directly connected routes corresponding to the eight VLAN subnets. Test 10 verifies that the access control list is matching traffic correctly by examining the hit counters in show access-lists. Test 11 verifies the operational state of DHCP through show ip dhcp pool and show ip dhcp binding, confirming that pools are active and that wireless clients have successfully obtained leases.

The fourth set of tests verifies wireless connectivity. Test 12 confirms that a wireless laptop has obtained an IP address from the correct DHCP pool via the ipconfig command. Test 13 verifies that the wireless laptop can ping a wired PC in the same VLAN. Test 14 verifies that the wireless laptop can ping a PC in a different VLAN through the router. Test 15 confirms that the access control list applies equally to wireless traffic by verifying that a wireless laptop in the College of Humanities cannot ping a PC in the College of Engineering.

Finally, Test 16 verifies the design's redundancy by simulating a failure of the active uplink between an access switch and Dist-SW-1. After disconnecting the cable, the dual-homed access switch reconverges through RSTP and activates its backup uplink to Dist-SW-2. Continued ping

traffic experiences a brief interruption of less than one second before resuming normally, demonstrating that the design tolerates the failure of a distribution switch or its primary link without operator intervention.

All sixteen tests produced the expected results, providing strong evidence that the design satisfies the segmentation, routing, security, redundancy, and operational requirements specified in the project brief.

8. Budget Proposal

The total estimated cost of the network design is $142,616, representing 71.3% of the $200,000 budget. The remaining $57,384 (28.7%) provides a substantial contingency margin, allowing for unforeseen costs such as additional cabling for buildings with non-standard layouts, replacement of equipment damaged during installation, or scope adjustments requested by individual colleges during deployment. The conservative pricing reflected in this budget assumes a procurement strategy that includes Cisco-certified refurbished equipment from authorized resellers for switches, access points, and IP telephony, a practice widely accepted in academic and public-sector procurement to reduce capital expenditure while maintaining vendor support and warranty coverage.

Table 2: Itemized budget summary with vendor sources.

Category	**Item**	**Qty**	**Unit ($)**	**Subtotal ($)**
Network Core	Cisco ISR 4331 Router	1	895	895
	Cisco ASA 5506-X Firewall	1	549	549
	Cisco 2960-48TT-L (Dist.)	2	219	438
Access Layer	Cisco 2960-48TT-L (Access)	63	219	13,797
Wireless	Cisco Aironet 3702i AP	25	99	2,475
Telephony	Cisco IP Phone 7821	642	117	75,114
Peripherals	HP LaserJet 3301fdw	32	639	20,448
Cabling	Cat 6 cable (1000ft rolls)	14	99.95	1,400
	Racks, patch panels (est.)	1	3,500	3,500
Services	Installation & configuration	1	16,000	16,000
	Maintenance + training (Y1)	1	8,000	8,000
Total				**142,616**

The largest single line item is IP telephony at $75,114 for 642 Cisco 7821 phones, accounting for 53% of the total budget. The access layer follows at $13,797 for 63 switches, made affordable by leveraging certified refurbished pricing of $219 per unit. Faculty desktop computers and laptops are explicitly excluded from this budget on the assumption that end-user computing is procured through separate per-college IT allocations, as is standard in enterprise procurement.

The substantial budget margin of $57,384 enables several discretionary enhancements that could be considered during the deployment phase. These include upgrading from refurbished to new-condition switches for the distribution layer to extend the warranty period, adding a Cisco Wireless LAN Controller for centralized AP management (approximately $2,500), or expanding the wireless coverage to 35 access points for higher-density deployment in lecture halls and shared spaces. Alternatively, the surplus may be retained as a true contingency fund to absorb unexpected

procurement variances or used to fund Year 2 maintenance after the initial year of vendor support expires.

9. Conclusion and Future Enhancements

This design satisfies all the requirements specified in the IS457 Design Project brief. It provides each of the eight colleges with its own VLAN and subnet, drawn from a coherent and scalable address plan based on the private 10.0.0.0/16 block. Inter-VLAN routing is implemented through a router-on-a-stick configuration on a Cisco ISR 4331. Redundancy is achieved through RSTP and EtherChannel between two distribution switches, with each access switch dual-homed to both. Security is implemented in depth across four layers: a perimeter firewall, inter-VLAN ACLs, switch-level port security, and device hardening on all Cisco equipment. Wireless coverage is integrated through 25 access points connected to college-specific VLANs, with DHCP services hosted on the central router. The total cost of $192,850 fits comfortably within the $200,000 budget, leaving a meaningful contingency margin.

Several enhancements could improve the design in subsequent deployment phases. First, migration from WPA2-PSK to WPA2-Enterprise with 802.1X authentication would replace the shared pre-shared keys with per-user credentials authenticated against a central directory service, dramatically improving wireless security and accountability. Second, the addition of a Cisco Wireless LAN Controller would centralize the management of the 25 access points, enabling unified policy configuration, simplified firmware updates, and seamless client roaming between APs.

Third, the access layer could be migrated to stackable Cisco Catalyst 2960-X switches or to the more modern Catalyst 9200 series, which support StackWise interconnect technology. This would allow each college's stack of switches to be managed as a single logical device, reducing operational complexity. The cost of this upgrade was estimated during the design process at approximately $208,000 for switches alone, which exceeds the project budget; however, it would be a worthwhile second-phase investment when the institution's IT spending allows.

Fourth, the introduction of a centralized network monitoring infrastructure based on Simple Network Management Protocol (SNMP), syslog, and NetFlow would provide visibility into network health, performance, and security events. Tools such as Cisco Prime Infrastructure or open-source alternatives such as LibreNMS could aggregate this telemetry into actionable dashboards for the IT staff. Fifth, virtual private network (VPN) services through the existing ASA firewall would extend secure remote access to faculty members working from home or while traveling, with IPsec or SSL VPN tunnels terminating at the firewall.

Beyond these technical enhancements, the design also offers a foundation for organizational maturation in network operations. Establishing a formal change management process, documenting standard configurations in a configuration management database, and adopting a regular vulnerability scanning cadence would all complement the technical design with operational discipline. As the university grows and its dependence on network services increases, these processes become

increasingly critical.

In summary, the proposed design is technically sound, fits within the prescribed budget, and provides a solid foundation for the campus's networking needs. The defense-in-depth security posture, the redundancy mechanisms at the distribution layer, and the integration of wired and wireless connectivity into a unified policy framework collectively address the modern requirements of an academic enterprise network. Future enhancements outlined above offer a clear roadmap for evolving the design as the institution's needs grow.

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